

COP3330C Module 9 Programming Assignment 1: Applying the Adapter Pattern with Java Collections

In this assignment, you will solidify your understanding of Java collections, generics, and design patterns by creating a Java program that demonstrates the use of the Adapter pattern, along with other Java concepts such as the Diamond Operator, sorting a list of numbers, using the `remove()` method, and employing unbounded wildcards. Note that this assignment requires the submission of a source code file and a screenshot of output after executing the program.

Instructions:

- **Setup Your Project:**
 - Create a new Java class named `ConceptDemo`.
 - Write a `main()` method to implement all of the required tasks.
- **Apply the Adapter Pattern:**
 - Create an `ArrayAdapter` class that wraps an `Integer[]` and provides a `toList()` method that returns a new, mutable `List<Integer>` (implemented as an `ArrayList<Integer>`).
 - Create an `Integer[]` array with several elements (e.g., 5, 2, 9, 1, 4, 7).
 - **Note:** We are using an Object Adapter (rather than a Class Adapter). The `ArrayAdapter` contains an `Integer[]` as a component and adapts it to the `List<Integer>` interface instead of exposing the array directly.
- **Use the Diamond Operator:**
 - In the `toList()` method, create the `ArrayList<Integer>` using the Diamond Operator (`<>`).
 - Use `Collections.addAll()` to populate the list with the elements from the array.
- **Sort the List:**
 - In the `main()` method, call the adapter's `toList()` method and store the returned `List<Integer>`.
 - Print the list before sorting.
 - Use `Collections.sort()` to sort the elements of the list.
 - Print the sorted list to verify the results.
- **Use the `remove()` Method:**
 - Remove a specific element from the list by value (e.g., remove the value 4). Use `Integer.valueOf()` to avoid removing by index.
 - Print the list again to show the result after the removal.
- **Demonstrate Unbounded Wildcards:**
 - Create a method named `printList()` that accepts a `List<?>` parameter.
 - Call `printList()` from the `main()` method, passing the `List<Integer>` created by the adapter.
 - Use the `printList()` method to display the list's contents..
- **Testing:**
 - Your program should display labeled output showing:

- The list before sorting.
- The list after sorting.
- The list after removing the value 4.
- The list printed by calling the printList() method.

Program Output Example:

Before sorting: [5, 2, 9, 1, 4, 7]

After sorting: [1, 2, 4, 5, 7, 9]

After removing 4: [1, 2, 5, 7, 9]

Printing list using unbounded wildcard: [1, 2, 5, 7, 9]

Functional Requirements:

- Adapter Pattern: Correctly adapt an Integer[] array to a List<Integer>.
- Diamond Operator must be used for list creation.
- Sorting must be demonstrated using the Collections.sort() method.
- Remove an element from the list by specifying the value.
- Unbounded wildcard must be used in the printList() method.

Non-Functional Requirements:

- 4-line ID Header (top of the source file(s)), <= 80 columns each:
 - // FileName.java
 - // your name
 - // submission date
 - // one-line summary of program's purpose
- All lines <= 80 columns (including comments)
- Use of appropriate comments in your code to explain each major step.
- Correct indentation (minimum of 3 spaces per level)

Submission:

- Submit your ConceptDemo.java file to the GitHub repository.
- Submit a screenshot (.png or .jpg only) of the program output demonstrating the steps as shown above. The screenshot must be clear and legible and show only the output, do not submit an image of the entire IDE window or source code.
- Do not submit .class files.